

Two-Electrode Voltage Clamp

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Abstract

Two-electrode voltage clamp (TEVC) is a conventional electrophysiological technique used to artificially control the membrane potential (V_m) of large cells to study the properties of electrogenic membrane proteins, especially ion channels. It makes use of two intracellular electrodes—a voltage electrode as V_m sensor and a current electrode for current injection to adjust the V_m , thus setting the membrane potential at desired values and recording the membrane current to analyze ion channel activities. Here we describe the use of TEVC in combination with exogenous mRNA expression in *Xenopus* oocytes for ion channel recording.

Key words Voltage clamp, Membrane potential, Voltage electrode, Current electrode, *Xenopus* oocytes, Ion channel

1 Introduction

Ion channels of cell membranes may be gated by the membrane potential (V_m) and/or by specific chemicals. Those belonging to the superfamily of voltage-dependent ion channels are gated by the V_m directly, whereas others, classified as ligand-gated ion channels, are gated by their binding with specific chemicals called ligands. On the other hand, voltage-dependent ion channels can be subject to the modulation by chemical factors such as neurotransmitters, hormones, intracellular messengers or exogenous drugs; and some ligand-gated ion channels (such as NMDA receptors) are also affected by the change in V_m . Thus, to study the voltage dependent characteristics of ion channels readily, or to distinguish between the effects of V_m versus chemicals, an experimental procedure is needed to control the V_m (i.e., to change V_m in a desired pattern or set it at a desired level). This procedure, known as voltage clamp, was first designed by Cole and Marmont and improved by Hodgkin, Huxley and Katz for application to giant axons of squids as two-electrode voltage clamp (TEVC) in late 1940s (1). It utilizes two intracellular electrodes, one to monitor V_m and the other to inject

a current to adjust V_m to desired values (the injected current being equal to the membrane current). Although the use of TEVC is limited to giant axons or large cells such as skeletal muscle cells, much of our understanding of the basic biophysical properties of ion channels has been acquired by using this approach. It was based on this method in combination with experiments using extracellular electrodes that Neher and Sakmann developed the currently widely used single electrode patch clamp technique (2). Nowadays popular though the latter is, it's only suitable for clamping relatively small cells, but unfit for large cells; for the large currents would cause a significant voltage drop across the recording electrode that cannot be compensated to an acceptable extent. Thus, TEVC is still irreplaceable in voltage clamping large cells, especially *Xenopus* oocytes which are often used for exogenous expression of ion channels or receptors. The *Xenopus* oocyte is a convenient expression system widely employed to study the structure and function of ion channels and receptors using TEVC in combination with various molecular biological approaches, since its membrane has low expression of endogenous channels and receptors.

The basic principle of TEVC recording from an oocyte is schematically depicted in Fig. 1. V_m is monitored by connecting the voltage electrode (electrode 1) to the input of a voltage follower (A_1), which has very high input impedance and draws negligible

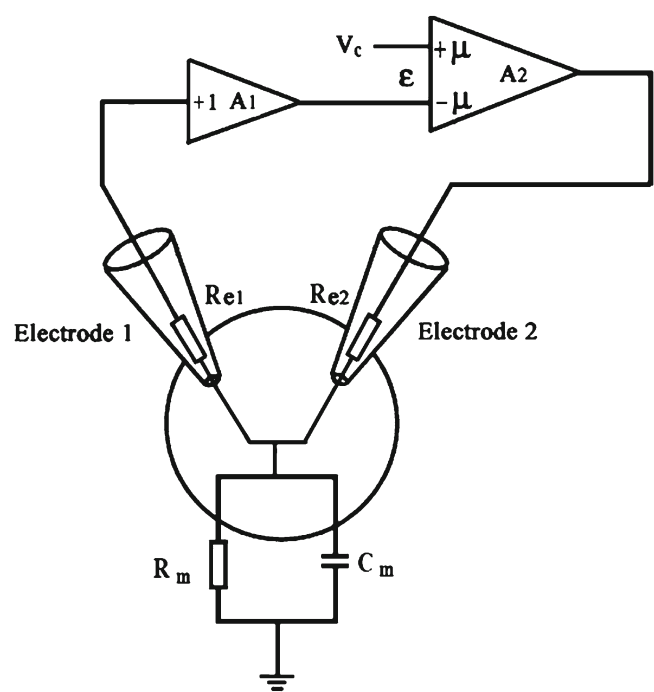


Fig. 1 Conventional two-electrode voltage clamp (TEVC) on an oocyte (see text for detail)

current from the cell. The output of A_1 , the potential at which is approximately equal to V_m , is connected to one input terminal of a clamping amplifier (A_2), which is a high gain differential amplifier and compares V_m with the voltage command signal (V_c) applied to the other input terminal. The resulting output voltage of A_2 forces a current proportional to the difference (ε) between V_m and V_c , usually by means of a voltage-controlled current source mechanism, to flow through the current electrode (electrode 2) into the cell. The direction of the current is decided by the polarity of the voltage at the output of A_2 , always reducing ε . As long as the gain (μ) of A_2 is sufficiently high ($\geq 10^3$), V_m can approach V_c very closely and thus achieves V_m clamp. Meanwhile, the current that passes through electrode 2 is measured (either upstream of electrode 2 or at the virtual grounding circuit) as the membrane current, since it counterbalances the V_m deviation the membrane current would have caused (3, 4).

Below we exemplify the TEVC procedures with the recording of Kv7.2/7.3 voltage-gated potassium channel currents.

2 Materials

All solutions are prepared using analytical grade reagents and ultra-pure water (resistivity 18 MΩ·cm). Stock solutions and working solutions are stored at 4°C, and are used within 1 month. Longer time of storage is not advised. All waste materials, especially the discarded glass electrodes that may cause injury, should be securely disposed of in accordance with local laboratory regulations.

2.1 Equipment and Materials

1. TEVC amplifier (e.g., GeneClamp 500B, Molecular Devices).
2. Digitizer (e.g., Digidata 1440A, Molecular Devices).
3. TEVC software (e.g., Clampex 10, Molecular Devices).
4. Micromanipulators (e.g., M3301, World Precision Instruments).
5. Faraday cage (custom-made).
6. Vibroisolating table (e.g., Newport).
7. Stereomicroscope (e.g., PZMIII-BS, World Precision Instruments).
8. Perfusion system (e.g., ALA-VM8, ALA Scientific Instruments).
9. Perfusion chamber (custom-made).
10. Aspiration system (e.g., vacuum pump, ZT-I, Shaoxing Medical Apparatus).
11. Micropipette puller (e.g., P-97 Micropipette Puller, Sutter Instruments).
12. Glass capillaries (e.g., B10024F, VitalScan Scientific Instruments).

13. Laboratory microwave oven.
14. Water bath.
15. Microinjector (e.g., Nanoliter Injector 2000, World Precision Instruments).

2.2 Preparation of External Solutions

1. Oocyte Ringer 2 solution (OR-2): 82.5 mM NaCl, 2 mM KCl, 1 mM MgCl_2 , 5 mM HEPES, pH 7.4. Weigh 4.82 g of NaCl and 1.19 g of HEPES and transfer them to a 1 L glass beaker, add 800 mL of ultrapure water, add 2 mL of 2 M KCl and 1 mL of 1 M MgCl_2 stock solutions, mix and dissolve them completely with the aid of a magnetic stirrer. Adjust the pH of OR2 to 7.4 with NaOH. We usually prepare 1,000 mL of this solution using a volumetric flask (see Notes 1 and 2).
2. ND96: 96 mM NaCl, 1 mM KCl, 1.8 mM CaCl_2 , 1 mM MgCl_2 , 5 mM HEPES, pH 7.4. First prepare 1 L of ND96 stock (5 $\times$): weigh 28.05 g of NaCl, 372.75 mg of KCl, 998.91 mg of CaCl_2 , 1,016.5 mg of MgCl_2 , 5,957.5 mg of HEPES, respectively, and put them into a 1 L beaker. Add water and stir to dissolve them completely, and then bring the volume to 1,000 mL with water. The stock solution is stored at 4°C. Prepare the final solution before use by dilution; adjust the pH to 7.4 with NaOH (5) (see Note 3).

2.3 Preparation of Electrodes

1. Preparation of Ag/AgCl wires. Immerse silver wires (4–5 cm long and 0.25 mm thick) in chlorine bleach with only a short end (less than 1 cm) of each wire left in the air, until the immersed part becomes uniformly blackened (see Note 4).

For making Ag/AgCl wires to be used in the voltage electrode and the current electrode, straight silver wires are used; for Ag/AgCl wires used for grounding the bath solution, it is better to coil part of the silver wire before coating, with narrow intervals left between adjacent turns of the coil. An alternative for grounding the bath solution is to use Ag/AgCl pellets (see Note 5).

2. Preparation of glass electrodes. The instructions below refer to the popular P-97 puller from Sutter Instruments; if using a different puller, refer to the instruction manual. Install a borosilicate glass pipette onto a P-97 puller, and run RAMP TEST to determine the heat value (RAMP value) for melting the glass (see Note 6). Repeat the determination and get an average RAMP value. Install a glass pipette of the same type and lot number, set the parameters of the puller as follows: HEAT = RAMP + 15, PULL = 150, VELOCITY = 100, TIME = 150, PRESSURE = 500 (see the P-97 puller manual for details) (6), and pull the pipette into two symmetric glass electrodes. Trim the tips of the glass electrodes using a pair a

fine scissors if needed (see Note 7), and store them in an electrode container.

3. Prepare 100 mL of 3 M KCl in a beaker, add 1 g of agarose, and microwave the mixture until the agarose is dissolved. Place the beaker containing agarose–KCl solution into a 80°C water bath to avoid agarose gelation. Warm up the syringe and its needle, with which the agarose–KCl solution is then injected into the glass electrode (see Note 8). Store the filled glass electrodes in a container (see Note 9).
4. Install a filled glass electrode onto the electrode holder (see Note 10), which, with Ag/AgCl wire pre-installed, has been secured to the headstage, the latter having been installed on a micromanipulator. Adjust the position of the tip of the glass electrode using the micromanipulator.
5. Measure the resistance of the electrode after its tip has been lowered into the bath solution with the micromanipulator. The electrode resistance should be in the range of 0.5–1 M Ω (see Note 11). If the electrode resistance is not within this range, adjust the parameters of the puller or modify the length of the tip for truncation.

2.4 Preparation of *Xenopus* Oocytes with Channels or Receptors Expressed

1. Preparation of *Xenopus* oocytes

Anesthetize the clawed toad by freezing it with ice until it is immobilized. Make a small longitudinal incision (0.5 cm) about 0.5 cm apart from the midline at the lower abdomen, and cut several ovarian lobes off. Stitch up the incision with 5–0 gauge suture. Position the toad in a basin until it regains consciousness and then place it back to water.

Cut the ovarian lobes into cubes of 3–4 mm in size, and digest them with OR2 containing collagenase (2 mg/mL) in a shaker at room temperature for 1.5–2 h. Stop digestion when most oocytes have been released and there are no connective tissues and capillaries left on the surface of the oocytes. Wash the oocytes repeatedly with OR2, and then wash them with ND96 three times. Select the large, healthy oocytes under a dissecting microscope, and keep them in ND96 at 18°C for a few hours before RNA injection.

2. Injection of mRNA into oocytes

Fill a glass needle (which has a long tip truncated to 20–30 μ m in OD at the tip opening) with corn oil, and install it onto the injector (Nanoliter Injector 2000, WPI). Drop 1–3 μ L of mRNA solution onto a piece of clean parafilm, and draw it into the glass needle with the injector under the microscope. Make sure that the mRNA is free from RNase contamination and no air has been drawn into the needle.

Transfer the oocytes together with ND96 solution onto a Petri dish with meshes at its bottom with a sucking glass pipette

(its opening fire-polished). Position the tip of the glass needle close to the surface of the oocyte and then impale it by using a micromanipulator. If the oocyte is in a good state, a resistance can be felt when the tip of the needle touches the oocyte membrane, and a “dimple” can be seen at the oocyte membrane before the needle can be inserted into the oocyte (in contrast, with unhealthy oocytes the impalement is much easier). Inject 46 nL of KCNQ2 or KCNQ3 mRNA solution (i.e., 3 ng mRNA) into each oocyte (the amount of injection of mRNA for other channels or receptors varies with the degree of difficulty in their expression and the status of the oocytes). After injection of an oocyte, wait for 1–2 s and then withdraw the needle carefully. Before turning to inject another oocyte, do an out-of-oocyte injection with the needle tip in the medium to make sure of the accuracy of next oocyte injection. After finishing the injection of all the oocytes, transfer them into ND96 solution containing gentamicin (50 mg/L) and sodium pyruvate (2.5 mM) and store at 18°C in an incubator. After 1–2 days TEVC experiments can be performed.

3 Methods

(All the following operations are carried out at room temperature).

3.1 *Checkup of the Recording System*

1. Check the perfusion system making sure that there is no blockage within it. Load solutions for the experiment, adjust the perfusion rate to about 2 mL/min (or to the value required for the specific experiment to be performed).
2. Check the aspiration system and drain tubing to make sure that they work properly so that the surface of the bath solution is stable.
3. Check the arrangement of the voltage headstage and current headstage installed on the micromanipulators, and make sure that they are positioned at a wide angle of over 90° (see Note 12).
4. Check the shielding and grounding of the apparatus around the input of the headstages to avoid introduction of external interferences, especially the power line frequency pickup (see Note 13).
5. Adjust the dissecting microscope so that the bath chamber to accommodate the oocyte is well within the visual field and clearly seen.
6. Turn on the system: turn the power of the digitizer on, and then turn on the power of the voltage clamp amplifier.
7. Configure the acquisition software and edit the voltage command protocol. For users of Clampex: open the software, click

“File” → “Set Data File Names” to designate the file name format and the folder to store the data to be recorded. Click “Configure” → “Digitizer” to change the configuration status from “Demo” to the digitizer being used. Create/edit the voltage command protocol for stimulation of an oocyte.

3.2 Load the Oocyte

Transfer a *Xenopus* oocyte expressing exogenous ion channel(s) and/or receptor(s) of interest gently to the bath chamber using a sucking pipette.

3.3 Observation and Recording of Currents

Steps described below are specific to Geneclamp 500B amplifier and the Clampex software; however, other TEVC hardware and software packages will have analogous steps/commands.

1. Lower the tips of the voltage electrode and current electrode into the bath using the micromanipulators, and position the electrode tips near (but not in contact with) the surface of the oocyte.
2. Zero offset potentials: press the buttons “ZERO V1” and “ZERO V2” on the front panel of the amplifier, and the “DC METERS” will read 0.
3. Measure electrode resistances: Press buttons “R1” and “R2” respectively, and the resistance of the electrodes will be displayed. The most appropriate values are within 0.5–1 M Ω (see Note 11).
4. Impale the oocyte: Under the dissecting microscope, press the tips of the electrodes against the surface of the oocyte gently using the micromanipulator, and impale the oocyte by further advancing the electrode tip slowly until it pops in (first do cell impalement with the voltage electrode and then with the current electrode) (see Note 14) (Fig. 2) (7).

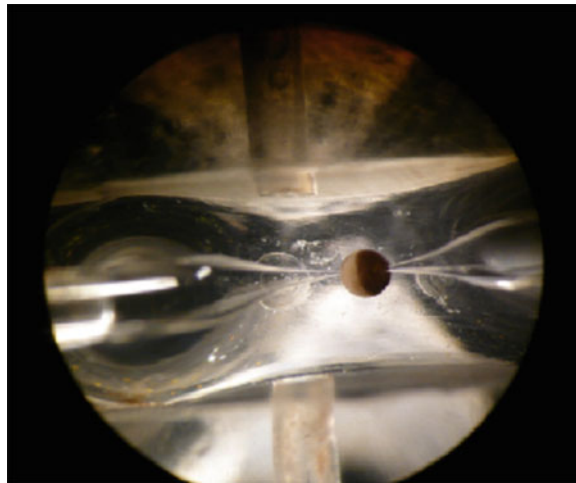


Fig. 2 The visual field of the stereomicroscope showing the *Xenopus* oocyte in the bath chamber and the two glass electrodes

“DC METERS” will display the resting membrane potential (RMP) of the oocyte; for example, the RMP of oocytes with overexpressed exogenous Kv7.2/7.3 heteromeric voltage-gated K⁺ channels is around -50 mV.

5. Position a grounded metal shield between the voltage and current electrodes if necessary (see Note 15).
6. Switch to voltage clamp mode: preset the holding potential at the RMP value (see Note 16). Then press button “VOLTAGE CLAMP.”
7. Gain and stability adjustment: Apply a series of test voltage pulses of -10 mV in amplitude and observe the current response flowing through the current electrode. Increase the GAIN of the clamping amplifier and the transients at the onset and the finish of the test pulses will sharpen up and even begin to oscillate. Then increase phase lag with the STABILITY control, and the oscillation will disappear. Repeat the operation, that is, increase the GAIN and the phase lag iteratively until the clamping amplifier operates at the minimum phase lag that permits the use of the maximum gain (see Note 17) (8). Then turn off the test pulses.
8. Observation of currents before formal recording: click button “View Only” to run the voltage command protocol and observe Kv7.2/7.3 mediated potassium currents (Fig. 3). If the leak currents are not small enough to be negligible compared with the active Kv7.2/7.3 currents, use “LEAK SUBTRACT” to remove the leak currents for better observation of the active currents.
9. Recording of currents: when the currents have become stable, use “RECORD” command to record the currents, and apply drugs to test their effects according to the experimental design. Full wash-out of one drug with the normal external solution such as ND96, which takes at least 2 min, is needed before testing another drug. When the recording of an oocyte is finished and another oocyte is to be recorded, press “STOP”

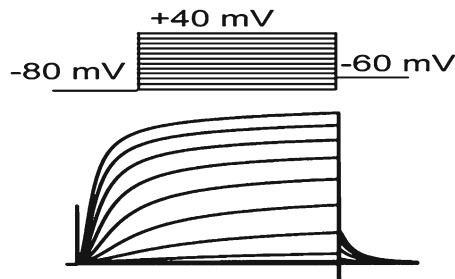


Fig. 3 Exemplary Kv7.2/7.3 mediated currents and the voltage clamp protocol used to induce the currents

to stop recording. Then press “SETUP” to stop voltage clamp and switch to setup mode, and then slowly withdraw the electrodes (see Notes 18 and 19). Repeat the aforementioned procedure to record currents from another oocyte.

3.4 Turning Off the Recording System and Cleaning the Setup

First close the software, then turn off the power of the digitizer and the amplifier respectively. Empty the remaining liquid in the perfusion system, and wash the system thoroughly with water in case its tubing(s) may be obstructed. Empty the waste liquid collected from the outlet of the vacuum pump.

4 Notes

1. OR2 is used to dissolve collagenase II for digestion of the layer of follicular cells surrounding the oocyte. It does not contain Ca^{2+} so as to avoid calcium overload in the oocyte during the digestion.
2. Only after all reagents are completely dissolved forming a clear solution by stirring the mixture should the pH adjustment be done.
3. The osmolality of normal external solutions for oocytes is ~ 189 mOsm/kg.
4. Silver wires can be well coated with AgCl layer after immersion in newly purchased or well sealed chlorine bleach for 20–30 min. Using outdated or poorly preserved bleach may take longer time or even produce no visible AgCl layer.
5. For recording large currents, a virtual ground circuit is generally used. Its voltage sensing electrode should be placed in the bath near the cell surface, so that the bath potential at the outside of the cell membrane can be clamped at zero without being affected by the large current flowing through the bath solution.
6. The result of the first RAMP test should be discarded because it often significantly deviates from those of the following tests due to some destabilizing factors.
The RAMP value should be determined for three times to get a mean value as reference for HEAT value setting.
7. This is because the opening of the electrode tips should not be too thin. Otherwise the electrodes would have a large resistance after being filled (see Note 11).
8. For filling the electrodes, the agarose–KCl solution should not occupy more than half the total electrode length in case the solution may spill out of the electrode and spoil the holder. Care should be taken not to have air bubbles left in the agarose between the electrode tip and the AgCl coated silver wire.

9. The filled glass electrodes can be used repeatedly. With a small amount of water at the bottom of the electrode container, the filled electrodes can be stored for a few weeks without the agarose inside the electrode drying up.
10. If the amount of agarose filled into the electrode is more than needed, the excess can be squeezed out of the electrode using a silver wire before installing the electrode onto the holder. Note that the upper end of the silver wire (uncoated with AgCl) should be well in contact with the input of the headstage.
11. If the resistance of the electrode is too large, the electrode tip can be truncated using a pair of fine scissors such as ophthalmic scissors. A current electrode of too large resistance will affect the passage of currents, while a voltage electrode of too large resistance will affect its frequency response and distort the V_m unless electrode capacitance neutralization (which may cause oscillation) is used.
12. Enlargement of the angle between of the two headstages (i.e., between the two electrodes to be installed) reduces their coupling capacitance, thus reducing the possibility of current oscillation.
13. Take care not to ground the case of the headstage (such as the HS-2A- $\times$ 1 LU for GeneClamp 500B amplifier) that uses a driven shield.
14. The tips of the two electrodes that have been inserted into the oocyte should be kept apart by some distance so as to reduce the error in V_m pick-up introduced by coupling between the two electrodes.
15. The grounded metal shield is used for reducing the coupling capacitance between the two electrodes, because this coupling capacitance is a destabilizing factor for TEVC. However, this shield should not contact the bath solution.
16. Presetting the holding potential at the RMP value can avoid the large current transient at the switch from SETUP mode to VOLTAGE CLAMP mode and large DC current after the switch. These currents may affect the cell becoming stable for recording.
17. The time constant (τ) for the change in V_m after a step voltage command is:

$$\tau = R_{e2} \times C_m / \mu$$

in which R_{e2} is the resistance of the current electrode, C_m is the membrane capacitance, and μ is the gain of the clamping amplifier (Fig. 1). It can be seen that increasing the gain of the clamping amplifier can make the V_m more close to V_c , thus

improving its fidelity and speed of response. Whereas, increasing phase lag decreases the high-frequency gain, and the clamping speed will be compromised when V_c jumps.

18. Generally, each glass electrode that has been filled and installed onto the headstage can be used repeatedly, i.e., on a few oocytes in succession. However, the electrode should not be used too many times for its resistance would become larger or unstable due to the blockade by intracellular components. Check the resistance of the electrode before reusing it on another oocyte. If its resistance value appears unstable or is out of the acceptable range, replace it with another electrode.
19. After recording large currents or after long time of use, the AgCl coated silver wire will become white, indicating the exhaustion of the AgCl coat. This may occur either at the grounding silver wire or at that connected to the headstage input, depending on the current direction. Take the silver wire off, rinse it, clean it carefully with tissue paper and then with fine sandpaper, and re-chloride it.

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